

16.1

$$a. \hat{Z} = \tilde{\gamma}_1 + \tilde{\gamma}_2 DTIME = -0.2376 + 0.5311 \times 1 = 0.2935 \quad NO.$$

Date

logit 的 CDF

$$\Lambda(Z) = \frac{1}{1+e^{-Z}} = \frac{1}{1+e^{-0.2935}} = 0.5729$$

$$b. \text{Probit 估計為 } \hat{B}_1 + \hat{B}_2 DTIME = -0.0644 + 0.3000 \times 1 = 0.2356$$

$$P(AUTO=1) = \Phi(0.2356) \approx 0.5931$$

與 logit model 相比 Probit model 估計的機率較大，但相差不多。

$$c. \frac{\partial P}{\partial DTIME} = \frac{\partial \Lambda(Z)}{\partial Z} \cdot \frac{\partial Z}{\partial DTIME} = \Lambda(Z)[1-\Lambda(Z)] \cdot \tilde{\gamma}_2$$

$$Z = -0.2376 + 0.5311 \times 30 = 15.70$$

$$\Lambda(Z) = \frac{1}{1+e^{-15.70}} \approx 1$$

$$\text{marginal effect} \approx 1 \cdot [1-1] \cdot 0.5311 \approx 0$$

當 DTIME 多大時，機率已接近 1，marginal effect 已趨近 0。

$$\text{LPM marginal effect: } \frac{\partial P}{\partial DTIME} = \hat{B}_2 = 0.3214$$

LPM 給出固定的邊際效果 0.3214，但 logit 在極端值的邊際效果趨近 0，logit 更合理。

d. logit marginal effect:

$$Z = -0.2376 + 0.5311 \times (-50) = -26.7926$$

$$\Lambda(Z) = \frac{1}{1+e^{26.7926}} \approx 0 \Rightarrow \text{marginal effect} = \Lambda(Z)[1-\Lambda(Z)] \cdot \tilde{\gamma}_2 \approx 0$$

$$\text{Probit marginal effect: } Z = -0.0644 + 0.3000 \times (-50) = -15.0644$$

$$\frac{\partial P}{\partial DTIME} = \phi(Z) \cdot \hat{B}_2 \approx 0 \quad \because \phi(-15.0644) \approx 0$$

兩個 model 在極端值下的 marginal effect 均為 0，結果一致，顯示非線性

模型在邊界處的行為較合理。

16.4

$$a. z = -1.836 + 3.021 \times 1.5 = 2.6955$$

$$p = \Lambda(2.6955) = \frac{1}{1 + e^{-2.6955}} = 0.9368$$

b. 門檻值 = 0.5

a 題得 $p = 0.9368 \Rightarrow \hat{y} = 1$ ，而實際值 y 也等於 1 $\Rightarrow$ 預測正確。

$$c. y_1 = +1, y_2 = 2 \Rightarrow z_i = -1 + 2x_i$$

$$i=1, x=1.5, y=1$$

$$z = -1 + 2 \times 1.5 = 2, \Lambda(z) = \frac{1}{1 + e^{-2}} = 0.8808, y=1 \Rightarrow \text{有 } 0.8808$$

$$i=2, x=0.6, y=1$$

$$z = -1 + 2 \times 0.6 = 0.2, \Lambda(0.2) = \frac{1}{1 + e^{-0.2}} = 0.5498, y=1 \Rightarrow \text{有 } 0.5498$$

$$i=3, x=0.7, y=0$$

$$z = -1 + 2 \times 0.7 = 0.4, \Lambda(0.4) = \frac{1}{1 + e^{-0.4}} = 0.5987, y=0 \Rightarrow \text{有 } 1 - 0.5987 = 0.4013$$

$$\text{likelihood function: } L = \prod_{i=1}^3 \Lambda(z_i)^{y_i} [1 - \Lambda(z_i)]^{1-y_i}$$

$$\text{log-likelihood function: } \ln L = \sum_{i=1}^3 [y_i \cdot \ln(\Lambda(z_i)) + (1-y_i) \cdot \ln[1 - \Lambda(z_i)]]$$

$$\ln L = \ln 0.8808 + \ln 0.5498 + \ln 0.4013 = -1.6378$$

$\Rightarrow$ 最大似然估計值 = -1.612 較大，符合 MLE 必定使 log-likelihood 達到最大的定義

d. True, 每個 $[\Lambda(z_i)]^{y_i} [1 - \Lambda(z_i)]^{1-y_i}$ 均為機率，都介於 0~1 之間，相乘後仍

於 0~1 之間。#

e. True, $\ln L = \sum_{i=1}^N \ln(\text{機率值})$ ，因每個機率值均介於 0~1 之間，故

$\ln(\text{機率}) < 0$ ，全部加總後仍為負值。#

16.18

a.

```
> lpm <- lm(delinquent ~ lvr + ref + insur + rate + amount + credit + term + arm,
+ data = lasvegas)
> coeftest(lpm, vcov = vcovHC(lpm, type = "HC1"))

t test of coefficients:

            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.68849128 0.22850643  3.0130 0.0026525 **
lvr          0.00162385 0.00067523  2.4049 0.0163601 *
ref         -0.05932367 0.02402561 -2.4692 0.0137098 *
insur       -0.48158489 0.03036944 -15.8575 < 2.2e-16 ***
rate        0.03437614 0.00981944  3.5008 0.0004845 ***
amount      0.02376801 0.01445089  1.6447 0.1003398
credit      -0.00044190 0.00020728 -2.1319 0.0332620 *
term        -0.01261951 0.00355596 -3.5488 0.0004051 ***
arm         0.12832392 0.02769325  4.6338 4.07e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

term 預期應為正,年期愈長風險會愈大,除了 term 符號有疑問外,其他變數符號均符合經濟直覺。
#

b.

```
Call:
glm(formula = delinquent ~ lvr + ref + insur + rate + amount +
  credit + term + arm, family = binomial(link = "logit"), data = lasvegas)

Coefficients:
(Intercept) 1.665303 1.992157 0.836 0.403195
lvr          0.013519 0.008199 1.649 0.099164 .
ref         -0.525064 0.230491 -2.278 0.022725 *
insur       -3.122105 0.216906 -14.394 < 2e-16 ***
rate        0.308656 0.082771  3.729 0.000192 ***
amount      0.217606 0.113769  1.913 0.055786 .
credit      -0.003564 0.001933 -1.844 0.065232 .
term        -0.133037 0.034597 -3.845 0.000120 ***
arm         1.405649 0.371355  3.785 0.000154 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 998.03  on 999  degrees of freedom
Residual deviance: 666.29  on 991  degrees of freedom
AIC: 684.29

Number of Fisher Scoring iterations: 5
```

所有變數符號均與 LPM 完全一致,顯著性也大致相同,兩個 model 結論穩健一致。

c.

```
> obs_500_1000 <- lasvegas[c(500,1000),]
> View(obs_500_1000)
> pred_lpm <- predict(lpm, newdata = obs_500_1000)
> pred_logit <- predict(logit, newdata = obs_500_1000, type = "response")
> data.frame(Observation = c(500, 1000),
+           Actual       = lasvegas$delinquent[c(500, 1000)],
+           LPM_pred     = pred_lpm,
+           Logit_pred   = pred_logit)
  Observation Actual LPM_pred Logit_pred
500          500     0 0.1827828 0.1335848
1000         1000     0 0.5785296 0.6265535
```

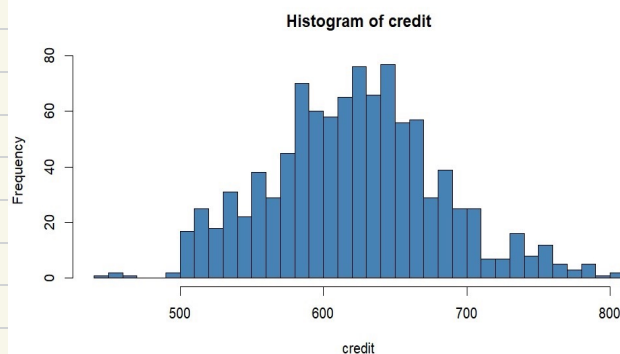
第500筆數據中,兩個 model 預測的違約機率都很低(約13%~18%),實際亦未違約,預測正確。
第1000筆數據中,兩個 model 預測的違約機率都超過 0.5 (約58%~63%),實際未違約,兩個 model 均預測錯誤。
而 logit 的預測值較端大(更靠近 0 或 1),LPM 較保守

d.

```

> base_case <- data.frame(
+   lvr = 80, ref = 0, insur = 0, rate = 8,
+   amount = 2.5, term = 30, arm = 0
+ )
> credit_vals <- c(500, 600, 700)
> results_d <- lapply(credit_vals, function(cr) {
+   df <- base_case; df$credit <- cr
+   data.frame(
+     credit      = cr,
+     LPM_prob    = predict(lpm, newdata = df),
+     Logit_prob  = predict(logit, newdata = df, type = "response")
+   )
+ })
> do.call(rbind, results_d)
  credit LPM_prob Logit_prob
1     500 0.5532956 0.4967030
11    600 0.5091060 0.4086308
12    700 0.4649165 0.3260580

```



兩個模型都顯示信用分數越高，違約機率越低，方向一致
 LPM 每增加 100 分，機率下降約 0.044 (固定)
 logit 每增加 100 分，機率下降幅度逐漸縮小，反映非線性特性
 整體而言 logit 預測的機率比 LPM 低

e.

```

> #e小題
> lpm_me_credit <- coef(lpm)["credit"]
> results_e <- lapply(credit_vals, function(cr) {
+   df <- base_case; df$credit <- cr
+   p <- predict(logit, newdata = df, type = "response")
+   me <- p * (1 - p) * coef(logit)["credit"]
+   data.frame(credit = cr, Logit_ME = me, LPM_ME = lpm_me_credit)
+ })
> do.call(rbind, results_e)
  credit Logit_ME LPM_ME
1     500 -0.0008910620 -0.0004418953
11    600 -0.0008613439 -0.0004418953
12    700 -0.0007832567 -0.0004418953

```

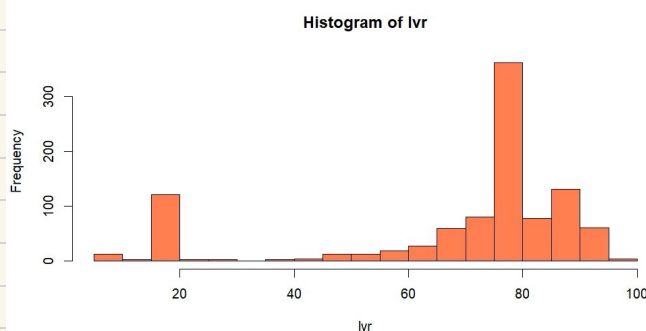
LPM：邊際效果固定為 -0.000442，不隨 CREDIT 變化
 Logit：邊際效果隨 CREDIT 升高而縮小，代表信用分數已經很高時，再提高一分的效果遞減
 Logit 的邊際效果在 CREDIT=500 時最大，比 LPM 大約兩倍，更能反映低信用分數者的風險敏感度

f.

```

> #f小題
> hist(lasvegas$lvr, main = "Histogram of lvr",
+   xlab = "lvr", col = "coral", breaks = 30)
> lvr_vals <- c(20, 80)
> results_f <- lapply(lvr_vals, function(lv) {
+   df <- base_case; df$lvr <- lv; df$credit <- 600
+   data.frame(
+     lvr      = lv,
+     LPM_prob = predict(lpm, newdata = df),
+     Logit_prob = predict(logit, newdata = df, type = "response")
+   )
+ })
> do.call(rbind, results_f)
  lvr LPM_prob Logit_prob
1   20 0.4116749 0.2349087
11  80 0.5091060 0.4086308

```



兩個模型都顯示 LVR 越高，違約機率越高，方向一致
 LPM 差距：0.5091 - 0.4117 = 0.0974
 Logit 差距：0.4086 - 0.2349 = 0.1737
 Logit 顯示 LVR 從 20 升到 80 的影響更大，LPM 低估了貸款成數的風險影響
 Logit 在低 LVR 時預測機率較低 (0.23)，更符合直覺：貸款成數低的人風險本來就小

9

```
> #g小題
> lpm_accuracy <- mean(ifelse(fitted(lpm) >= 0.5, 1, 0) == lasvegas$delinquent)
> logit_accuracy <- mean(ifelse(fitted(logit) >= 0.5, 1, 0) == lasvegas$delinquent)
> cat("LPM Accuracy: ", round(lpm_accuracy, 4), "\n")
LPM Accuracy: 0.858
> cat("Logit Accuracy:", round(logit_accuracy, 4), "\n")
Logit Accuracy: 0.854
```

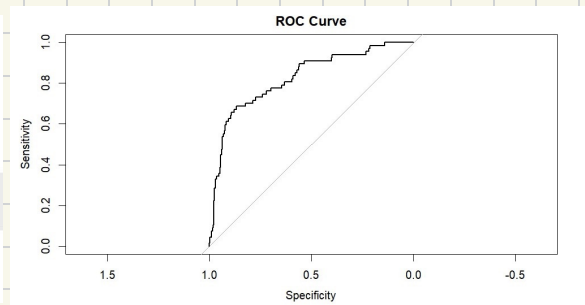
兩個模型準確率非常接近，僅差距 0.4%
LPM 略優於 Logit，但差距微小，無實質差異。

h.

```
> test$pred_prob <- predict(logit_train, newdata = test, type = "response")
> test$pred_class <- ifelse(test$pred_prob >= 0.5, 1, 0)
> acc_05 <- mean(test$pred_class == test$delinquent)
> cat("Accuracy (threshold=0.5):", round(acc_05, 4), "\n")
Accuracy (threshold=0.5): 0.88
>
> roc_obj <- roc(test$delinquent, test$pred_prob)

Setting levels: control = 0, case = 1
Setting direction: controls < cases

> plot(roc_obj, main = "ROC Curve")
> cat("AUC:", auc(roc_obj), "\n")
AUC: 0.8253076
```



用 0.5 非為最佳門檻，理由為：

1. 違約率本身很低，用 0.5 門檻幾乎不會拒絕任何人，大多數人都會被預測為不違約。
2. 兩種錯誤的代價不對等，若放貸給違約者，銀行會損失本金，代價相對大；若拒絕好借款人，銀行損失利息收入，代價相對小。

更好的做法：

- 降低門檻（如用實際違約率約 15~20%）
- 用 ROC 曲線找使 sensitivity + specificity 最大的最佳門檻
- AUC = 0.825 代表模型區分能力良好，值得用更精細的門檻來發揮

結論：應降低門檻以提高對違約者的識別率，減少銀行的實際損失風險。